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Centrality Measures in Networks

Zagreb, November 2024

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1 Abstract

This paper investigates centrality measures for vertices such as degree centrality, closeness centrality, eigenvector centrality, and Katz centrality. It is shown that Katz centrality can be transformed into degree centrality and eigenvector centrality by considering limiting cases. Some centrality measures are related to matrix functions. The concept of centrality measures is extended to measures based on general matrix functions such as the exponential function. On various graphs, using Kendall's coefficient, it is shown that different centrality measures need not yield the same results. The impact of graph type on the correlation of certain centrality measures is also addressed; specifically, the Erdős–Rényi, Watts–Strogatz, and Barabási–Albert models are used to generate commonly used graphs. Conjectures are derived based on the conducted experiments, concerning the proportionality of certain variables in models with correlations of different centralities.

2 Introduction

How to determine the most important vertices in a given graph? How do we define the importance of a vertex in the first place? Many real-world problems can be modeled as networks that describe relationships and phenomena such as friendships in a group of people, disease spread in a small community, or information flow in a telecommunication network.

Intuitively, important vertices in a group of friends would be those individuals who are exceptionally popular; they have a large number of friends. For such centrality, we would count the neighbors of vertices.

In the case of disease outbreak in a group of people, main transmitters would be those who are close to almost everyone. Here we would measure the average distance from others.

For a network of telecommunication towers, we would assign importance to those towers that are most critical for transmitting information from one place to another; their absence would cause loss in speed. Based on this, it would be desirable to count the shortest paths and in how many of them a particular tower participates.

Of course, for all these problems it is possible to take some other approach, that is, to use a different centrality measure; define a mapping that assigns a real number to vertices and based on the ordering determine the greatest or most important one. How large the difference will be when using different centrality measures is the main question we address in this paper.

It is also interesting to observe more abstract examples of centrality measures and compare their results with more commonly used ones; what would centrality defined by the sin or log function look like?

3 Introductory Concepts

3.1 Matrix Representation of Undirected Graphs

For an undirected graph G with n vertices and m edges:

- **Adjacency Matrix** of graph $G(V, E)$ is given by $A = \{a_{ij}\}_{n \times n}$, where

$$a_{ij} = \begin{cases} 1 & \text{if vertex } v_i \text{ is adjacent to vertex } v_j \\ 0 & \text{otherwise} \end{cases}$$

- **Incidence Matrix** is given by $M = \{m_{i,j}\}_{n \times m}$, where

$$m_{ij} = \begin{cases} 1 & \text{when edge } e_j \text{ is incident with vertex } v_i \\ 0 & \text{otherwise} \end{cases}$$

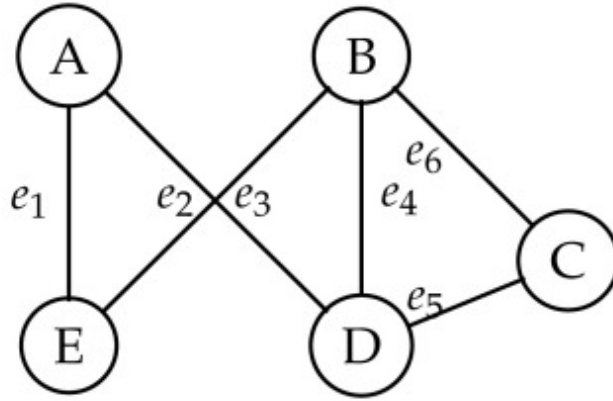


Figure 3.1

Figure 3.1 shows an undirected graph. The adjacency matrix and incidence matrices are given as:

$$A = \begin{pmatrix} 0 & 0 & 0 & 1 & 1 \\ 0 & 0 & 1 & 1 & 1 \\ 0 & 1 & 0 & 1 & 0 \\ 1 & 1 & 1 & 0 & 0 \\ 1 & 1 & 0 & 0 & 0 \end{pmatrix}, M = \begin{pmatrix} 1 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 1 & 0 & 1 \\ 0 & 0 & 0 & 0 & 1 & 1 \\ 0 & 1 & 0 & 1 & 1 & 0 \\ 1 & 0 & 1 & 0 & 0 & 0 \end{pmatrix}.$$

3.2 Matrix Representation of Directed Graphs

For a directed graph G with n vertices and m edges:

- **Adjacency Matrix** of graph $G(V, E)$ is given by $A = \{a_{ij}\}_{n \times n}$, where

$$a_{ij} = \begin{cases} 1 & \text{if vertex } v_i \text{ points to vertex } v_j \\ 0 & \text{otherwise} \end{cases}$$

- **Incidence Matrix** is given by $M = \{m_{i,j}\}_{n \times m}$, where

$$m_{ij} = \begin{cases} 1 & \text{if } v_i \text{ is the starting vertex of edge } e_j \\ -1 & \text{if } v_j \text{ is the starting vertex of edge } e_i \\ 0 & \text{otherwise} \end{cases}$$

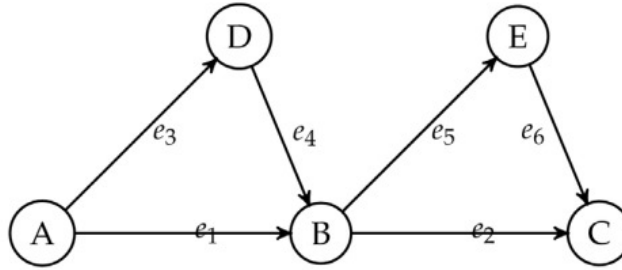


Figure 3.2

Figure 3.2 shows a directed graph. The adjacency matrix and incidence matrices are given as:

$$A = \begin{pmatrix} 0 & 1 & 0 & 1 & 0 \\ 1 & 0 & 1 & 1 & 1 \\ 0 & 1 & 0 & 0 & 1 \\ 1 & 1 & 0 & 0 & 0 \\ 0 & 1 & 1 & 0 & 0 \end{pmatrix}, M = \begin{pmatrix} 1 & 0 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 1 & 1 \\ 0 & 1 & 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 1 \end{pmatrix}.$$

3.3 Distance in a Graph

The shortest distance between two vertices v_i and v_j in a graph G is defined as the length of the shortest path between vertices v_i and v_j . Notation: $d(v_i, v_j)$.

The diameter of a graph $G = (V, E)$ is given as

$$\text{diam}(G) = \max_{v_i, v_j \in V} d(v_i, v_j).$$

The shortest distances in the directed graph Q in Figure 3:

$$d(D, C) = 2, \quad d(A, B) = 1 \quad \text{and} \quad d(D, D) = 0.$$

The distance matrix of a graph, denoted as D , is a square matrix $\{d(ij)\}_{n \times n}$, where $d(ij)$ equals the length of the shortest path from vertex v_i to vertex v_j .

Let us consider the undirected unweighted graph in Figure 3.1 as an example; the distance matrix for that graph is given as:

$$D = \begin{pmatrix} 0 & 2 & 2 & 1 & 1 \\ 2 & 0 & 1 & 1 & 1 \\ 2 & 1 & 0 & 1 & 2 \\ 1 & 1 & 1 & 0 & 2 \\ 1 & 1 & 2 & 2 & 0 \end{pmatrix}.$$

3.4 Perron-Frobenius Theorem

Theorem (Perron-Frobenius Theorem) Let A be an irreducible and non-negative $N \times N$ matrix with spectral radius $\rho(A) = r$. Then the following facts hold:

- The number $r \in \mathbb{R}^+$ is a real and positive eigenvalue of matrix A . We call it the **Perron–Frobenius eigenvalue**.
- The Perron–Frobenius eigenvalue r has **algebraic multiplicity 1**. In particular, the eigenspace associated with value r is one-dimensional.
- A has an eigenvector $\mathbf{w}$ associated with value r and all its components are positive. Furthermore, this is the unique eigenvector of matrix A that has all positive components up to multiplication by a positive scalar.

Of course, not all facts about the above theorem are stated, but the essential results are **the uniqueness of the eigenvector (up to non-zero scalar)** by which we determine which vertex is the center and the fact that the largest eigenvalue indeed exists and is real.

Note that we always work with connected graphs, which is the only thing that makes sense since vertices from one connected component have no effect on determining the centers of other components (so if we have k connected components, we apply the center-finding algorithm to each).

Theorem (Equivalence of Connectivity and Irreducibility) Let A be an $N \times N$ matrix and G be the associated graph. Then matrix A is irreducible if and only if G is a connected graph.

4 Centrality Measures

One of the key reasons for analyzing large networks is to determine which vertices are important or central in some sense. Which vertices have the greatest influence on others?

In this section, we proceed from the simplest to more complex methods of defining centrality in networks. We will consider the analysis successful if it answers the question *What exactly does this measure measure?* and *In a concrete situation, how do you decide which measure to use?*

All definitions are largely aided by intuition, which makes this branch of mathematics very rewarding.

4.1 Vertex Degree

Naturally, the first method that comes to mind is counting the vertices with which the observed vertex has a connection. It is immediately clear that this measures direct connectivity with other vertices, possibly neglecting important indirect connections.

Let A be the adjacency matrix of the observed graph G and $i \in V$ be a vertex of the graph. The degree of vertex i , denoted d_i , is defined as

$$d_i = \sum_{j=1}^n a_{ij} = e_i^T A e$$

For directed graphs, we want to track the direction of connectivity, so we naturally define two types of vertex degree for $i \in V$:

- in-degree: d_i^{in} = number of vertices pointing to vertex i
- out-degree: d_i^{out} = number of vertices that vertex i points to

Let us return to 3.2 and look at the corresponding vertex degrees:

- $d^{in} = (0, 2, 2, 1, 1)$
- $d^{out} = (2, 2, 0, 1, 1)$

4.2 Closeness Centrality

Unlike the previous method, here we also consider the indirect influence of other vertices on the observed vertex. More precisely, for a given vertex, we look at the average distance to all

other vertices, that is, for vertex $i \in V$ we define

$$l_i = \frac{1}{N-1} \sum_{i \neq j \in V} d(i, j)$$

where N is the number of vertices in graph G : Since we want to associate high importance of a vertex with a large associated number, we actually consider the reciprocal value

$$c_i = \frac{1}{l_i}$$

Let us again examine 3.1.

$$\begin{aligned} N = 5, l_1 &= \frac{2+2+1+1}{5-1} = \frac{3}{2}, c_1 = 0.66 \\ l_2 &= \frac{2+1+1+1}{5-1} = \frac{5}{4}, c_2 = 0.8 \\ l_3 &= \frac{2+1+1+2}{5-1} = \frac{3}{2}, c_3 = 0.66 \\ l_4 &= \frac{1+1+1+2}{5-1} = \frac{5}{4}, c_4 = 0.8 \\ l_5 &= \frac{1+1+2+2}{5-1} = \frac{3}{2}, c_5 = 0.66 \end{aligned}$$

so this measure would consider vertices 2 and 4 as the most important.

4.3 Eigenvector Centrality

For eigenvector centrality, the main idea is as follows: *A vertex is important if the important vertices point to it.* While in the first method we only counted the degree of a vertex, i.e., all neighbors were equally important to us, here we will try to *rank* the neighbors, so we expect better results than the simpler method - more on this in the next chapter.

Let A be an irreducible adjacency matrix of graph G and $n = |G|$. The goal is to arrive at a vector $\mathbf{x} \in (\mathbb{R}^+)^n$ from which we can conclude which vertex is most important by comparing its components.

The requirement is as follows:

The importance of a vertex is proportional to the sum of the importance of vertices pointing to it.

That is, for some $\lambda > 0$, we seek $\mathbf{x} = (x_1, \dots, x_n)$ such that

$$\forall i = 1, \dots, n, \quad x_i = \frac{1}{\lambda} \sum_{j=1}^n a_{ij} x_j \implies A\mathbf{x} = \lambda \mathbf{x}$$

Since we started with an irreducible matrix A , the Perron-Frobenius theorem guarantees us the existence of exactly such a vector and constant λ . This is the eigenvector associated with the largest eigenvalue λ .

Now that we know how important this eigenvector is, let us use it and iteratively arrive at the same ordering of vertices. Since A is symmetric, it is diagonalizable, so there exists a basis of its eigenvectors $\mathbf{v} = \{v_1, \dots, v_n\}$ where $Av_i = \lambda_i v_i$. WLOG assume that $\lambda_1 = \text{spr}(A) > 0$. Take initial $x^{(0)} = (0, \dots, 0)$ and define a sequence of iterations:

$$x^{(k)} = A^{(k)} x^{(0)}$$

Also, write $x^{(0)}$ in the basis $\mathbf{v}$

$$x^{(0)} = \sum_{i=1}^n \alpha_i v_i$$

Now we have

$$\begin{aligned} x^{(k)} &= A^{(k)} x^{(0)} = \sum_{i=1}^n \alpha_i A^{(k)} v_i = \sum_{i=1}^n \alpha_i \lambda_i^k v_i \\ \implies \frac{x^{(k)}}{\lambda_1^k} &= \alpha_1 v_1 + \sum_{i=2}^n \alpha_i \left(\frac{\lambda_i}{\lambda_1} \right)^k v_i \implies \left[\frac{\lambda_i}{\lambda_1} < 1, \text{ for } i > 1 \right] \implies \lim_{k \rightarrow \infty} \frac{x^{(k)}}{\lambda_1^k} = \alpha_1 v_1 \end{aligned}$$

Since the coefficients $\frac{1}{\lambda^k}$ do not change the ordering in the sequence members, we conclude that after a sufficiently large number of iterations, we arrive at the same ordering as in the vector $\alpha_1 v_1$ (or exactly the opposite, but then we just multiply by -1).

4.4 Katz Centrality

If centrality defined by vertex degree is one way to determine the center in a graph, and eigenvector centrality is another, then Katz centrality would be a combination of the two. The idea is to consider all paths that a vertex has with others, but penalize each longer path proportionally with some α constant. Using the already mentioned interpretation of the coefficients

of the matrix A^k , for the i -th coefficient we would calculate:

$$C(i) = \sum_{k=1}^{\infty} \sum_{j=1}^n \alpha^k (A^k)_{ji}$$

The interpretation is as follows: for each path of length k we reduce its importance by multiplying by α^k . Vectorially, by expanding into a series we obtain:

$$C = \left((I - \alpha A)^{-1} - I \right) e, \quad (1)$$

where e is a column vector of ones. Note that for this expression to be valid (to converge), we need $\alpha \in (0, \frac{1}{\lambda})$ where λ is the largest eigenvalue of the matrix A . The existence of such a positive eigenvalue follows from the Perron-Frobenius theorem. Furthermore, since we are only interested in the ordering between vertices, not the value itself given to each vertex, it is allowed to remove the identity matrix with which we subtract, so we have:

$$C = (I - \alpha A)^{-1} e, \quad (2)$$

4.5 Exponential Subgraph Centrality

Here we will estimate the importance of a vertex based on how much it participates in subgraphs of the given graph. By *participate*, we actually mean counting in how many closed walks the observed vertex participates. If it participates in many closed walks, we can interpret this as many subsets of vertices considering it important among themselves, not just globally.

There is a simple relationship between the adjacency matrix A and the number of closed k walks in a graph. Let us consider the matrix A^2 . Suppose that $[A^2]_{ij} = r > 0$. So we multiplied the i -th row of matrix A and its j -th column and got r matches of ones. Interpreted, r times we could reach some vertex and from it to the j -th vertex. So,

$$[A^2]_{ij} = \text{number of 2-walks from vertex } i \text{ to vertex } j$$

We conclude inductively

$$[A^k]_{ij} = \text{number of } k\text{-walks from vertex } i \text{ to vertex } j$$

Further, we define

$$\begin{aligned}\mu_k(i) &= [A^k]_{ii} = \text{local spectral moment of vertex } i \\ &= \text{number of } k \text{ closed walks from vertex } i\end{aligned}$$

Finally, we define the subgraph centrality of vertex $i \in V$

$$SC(i) = \sum_{k=0}^{\infty} \frac{\mu_k(i)}{k!} = [e^A]_{ii}$$

So, again guided by intuition, the more small subgraphs in which a vertex participates, the more important it is. Note that each k -walk can proceed in two directions starting from any of those k vertices, so we have a total of $2 \cdot k$ closed walks in such a situation and all of them are included in the matrix A^k . For this reason we penalize all these possible ways with $k!$, so as they increase, due to the dominance of the factorial, they become less and less important, which is the idea of the method. This idea can be enhanced by adding some positive coefficient β to each term of the infinite sum; usually β^k so that the coefficient enters the exponent, then we would have $SC_i(\beta) = [e^{\beta A}]_{ii}$. In some cases, the penalization with the factorial is omitted, and with a sufficiently small coefficient to ensure convergence of the series ($\beta \in (0, \frac{1}{\lambda})$, λ the largest eigenvalue), we get: $SC_i(\beta) = [(I - \beta A)^{-1}]_{ii}$. The first of the mentioned penalization methods gives the **exponential** form, while the second gives the **resolvent** form.

Let us consider the following graph and using the exponential form set $\beta = 1$.

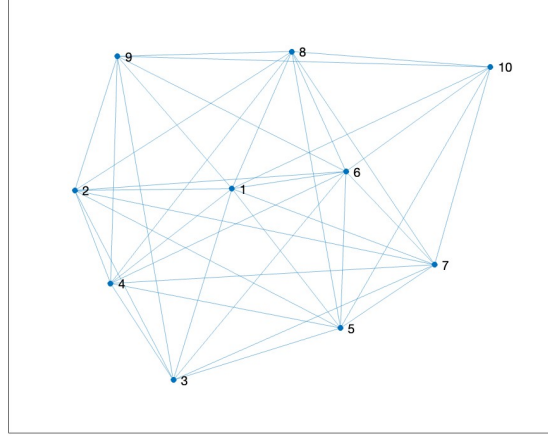


Figure 4.1

After calculation we obtain

$$e^A = \begin{bmatrix} 333.82 & 307.15 & 273.12 & 307.15 & 303.62 & 333.46 & 303.62 & 302.75 & 265.21 & 233.72 \\ 307.15 & 283.52 & 251.82 & 283.15 & 279.67 & 307.15 & 279.67 & 278.79 & 244.28 & 214.69 \\ 273.12 & 251.82 & 224.53 & 251.82 & 248.68 & 273.12 & 248.68 & 247.45 & 217.19 & 190.73 \\ 307.15 & 283.15 & 251.82 & 283.52 & 279.67 & 307.15 & 279.67 & 278.79 & 244.28 & 214.69 \\ 303.62 & 279.67 & 248.68 & 279.67 & 277.04 & 303.62 & 276.68 & 275.66 & 240.97 & 212.80 \\ 333.46 & 307.15 & 273.12 & 307.15 & 303.62 & 333.82 & 303.62 & 302.75 & 265.21 & 233.72 \\ 303.62 & 279.67 & 248.68 & 279.67 & 276.68 & 303.62 & 277.04 & 275.66 & 240.97 & 212.80 \\ 302.75 & 278.79 & 247.45 & 278.79 & 275.66 & 302.75 & 275.66 & 275.45 & 240.77 & 212.43 \\ 265.21 & 244.28 & 217.19 & 244.28 & 240.97 & 265.21 & 240.97 & 240.77 & 211.68 & 185.81 \\ 233.72 & 214.69 & 190.73 & 214.69 & 212.80 & 233.72 & 212.80 & 212.43 & 185.81 & 164.80 \end{bmatrix}$$

Looking at the diagonal of the matrix

$$SC = \begin{bmatrix} 333.82 & 283.52 & 224.53 & 283.52 & 277.04 & 333.82 & 277.04 & 275.45 & 211.68 & 164.80 \end{bmatrix}$$

we conclude that vertices 1 and 6 have the greatest importance according to this measure.

4.6 Total Communicability Centrality

The interpretation is exactly the same as for Katz centrality; the difference is in the way of penalizing terms of the form $(A^k)_{ji}$. We set for a fixed $\alpha \in (0, \infty)$ the penalization factor $\frac{\alpha^k}{k!}$

and by expanding into a series we obtain:

$$TC(\alpha) = e^{(\alpha A)} e$$

where e is a column vector of ones, and TC is the vector of corresponding centrality values.

4.7 Betweenness Centrality

If in some graph $G = (V, E)$ we use edges E to model communication channels between vertices V , then it is natural to ask which vertices have the greatest role in transmitting information; which vertices would we say are the most important intermediaries.

The calculation of this centrality is based on counting shortest paths: for some $v \in V$, we assign to it:

$$B(v) = \sum_{s \neq v \neq t} \frac{\sigma_{st}(v)}{\sigma_{st}}$$

where σ_{st} is the number of shortest paths from vertex s to vertex t and $\sigma_{st}(v)$ is the total number of such paths where v is an intermediary.

More complex algorithms are needed to calculate this centrality and typically modifications of the Floyd-Warshall algorithm (cubic complexity) are used, although for special structures such as sparse matrices that model weakly connected graphs, more efficient algorithms exist (Brandes' algorithm).

4.8 Graph Generation Models

In order to conduct a comparison of different centrality measures, we use three different types of graphs depending on the model from which they originated: Erdős–Rényi, Watts–Strogatz, and Barabási–Albert.

4.8.1 Erdős–Rényi Model

In this model, a graph is created completely randomly, i.e., each vertex randomly chooses its neighbors with a given probability p . Networks generated by this model consist of vertices with roughly equal degree, and such networks rarely have outliers.

4.8.2 Watts–Strogatz Model

This model generates a graph that has a very small average distance between vertices, but it is also characterized by a high clustering coefficient. So, there is a correlation between small average distance and strong groupings.

4.8.3 Barabási–Albert Model

Graphs obtained by this model fall into the category of "scale-free" graphs. The degree distribution of vertices in a "scale-free" graph follows an exponential distribution, i.e., most vertices have a small number of neighbors, yet there are a few vertices with large degree in the graph. An essential characteristic of this model is *preferential attachment*. New vertices in the network have a higher probability of connecting to vertices that already have many connections (the rich get richer).

5 Theoretical Comparison of Measures

After studying various centrality measures, it is natural to ask which measure is best to use. It is not obvious whether two centrality measures will give the same ordering of vertices in a particular network, and whether there is an optimal measure to use on a given set of data.

5.1 Coefficients and Their Impact on Centrality

Centralities such as Katz or subgraph centrality (whether exponential or resolvent) depend on the choice of coefficients in their definition; they will converge to already known ones such as eigenvector centrality or degree centrality.

The main idea in the following proofs is to exploit the fact that only the relationship between vertices matters, not the value associated with each vertex. Thus, transformations that preserve ordering can be applied to centrality functions because the final result does not change.

5.1.1 Exponential Subgraph Centrality

Theorem: Let $G = (V, E)$ be a connected undirected graph with adjacency matrix A . Let $SC_i(\beta) = [e^{\beta A}]_{ii}$ be the exponential subgraph centrality of vertex i and SC the associated vector. Then,

- (i) $\beta \rightarrow 0^+$ implies that the ordering given by $SC(\beta)$ converges to the ordering given by C_d ; the degree centrality vector;
- (ii) $\beta \rightarrow \infty$ implies that the ordering given by $SC(\beta)$ converges to the ordering given by C_{ev} ; the eigenvector centrality vector

Proof: The Taylor expansion of function $SC_i(\beta)$ is given by:

$$\sum_{k=0}^{\infty} \frac{\beta^k [A^k]_{ii}}{k!}.$$

The value assigned to index $k = 1$ is the coefficient β multiplied by 0 since graphs have no loops (a path of unit length that starts and ends at the same vertex), while at $k = 2$, the coefficient $[A^2]_{ii}$ carries the interpretation of the number of vertices connected to vertex i ; that is, the degree of vertex i , $C_d(i)$; do not forget that it is multiplied by $\beta^2/2$.

A function that clearly does not change the ordering because it is strictly increasing is:

$$f(x) = \frac{2!}{\beta^2} [x - 1]$$

Substituting $SC_i(\beta)$, we get:

$$\frac{2!}{\beta^2} [SC_i(\beta) - 1] = \mathbf{C}_d(i) + O(\beta)$$

By letting $\beta \rightarrow 0^+$, we get the claim of the first point.

To prove the second point, we use the fact that A is a Hermitian matrix, hence diagonalizable, so with $S = [q_1, q_2, \dots, q_n]$ unitary with orthonormal eigenvectors and D diagonal so that on it lie the associated eigenvalues $\lambda_1, \lambda_2, \dots, \lambda_n$, where the first eigenvalue is dominant from the Perron-Frobenius theorem, we have: $A = SDS^*$. Now, the Taylor expansion of $SC_i(\beta)$ takes a new form:

$$SC_i(\beta) = \sum_{k=1}^n e^{\beta \lambda_k} q_k(i)^2$$

We apply scaling:

$$g(x) = \frac{1}{e^{\beta \lambda_1}} x$$

And we get:

$$q_1(i)^2 + \sum_{k=2}^n e^{(\lambda_k - \lambda_1)\beta} q_k(i)^2$$

Since A is Hermitian, we know that the eigenvalues are real, so $\lambda_k - \lambda_1$ is a negative number, which with $\beta \rightarrow \infty$ gives us that $SC(\beta)$ tends to $q_1 \circ q_1$ (where $\circ$ is the Hadamard product). Since q_1 is a vector with positive values (Perron-Frobenius), taking square roots preserves the ordering, thus proving the second point of the theorem. □

5.1.2 Resolvent Subgraph Centrality

Recall that λ_1 is reserved for the largest eigenvalue of the adjacency matrix A .

Theorem: Let $G = (V, E)$ be a connected undirected graph with adjacency matrix A . Let $RC_i(\alpha) = (I - \alpha A)_{ii}^{-1}$ be the resolvent subgraph centrality of vertex i and RC the associated vector. Then,

- (i) $\alpha \rightarrow 0^+$ implies that the ordering given by $RC(\alpha)$ converges to the ordering given by $\mathbf{C}_d$; the degree centrality vector;

- (ii) $\alpha \rightarrow \frac{1}{\lambda_1} -$ implies that the ordering given by $RC(\alpha)$ converges to the ordering given by $\mathbf{C}_{ev}$; the eigenvector centrality vector

Proof: The Taylor expansion of function $RC_i(\alpha)$ is given by:

$$\sum_{k=0}^{\infty} \alpha^k [A^k]_{ii}.$$

We use similar argumentation, so we only state with which increasing function we compose:

$$f(x) = \frac{1}{\alpha^2} [x - 1]$$

We obtain:

$$\frac{1}{\alpha^2} [RC_i(\alpha) - 1] = \mathbf{C}_d(i) + O(\alpha)$$

By letting $\alpha \rightarrow 0^+$, the first point of the theorem follows.

By applying diagonal decomposition of matrix A , the new form is obtained:

$$RC_i(\alpha) = \sum_{k=1}^n \frac{1}{1 - \alpha \lambda_k} q_k(i)^2$$

We scale by $1 - \alpha \lambda_1$; we know that it is positive due to the condition $\alpha \in (0, \frac{1}{\lambda_1})$. We obtain:

$$q_1(i)^2 + \sum_{k=2}^n \frac{1 - \alpha \lambda_1}{1 - \alpha \lambda_k} q_k(i)^2$$

Now, $\alpha \rightarrow \frac{1}{\lambda_1} -$ is well-defined in the sense that expressions in the denominator cannot reach 0, so we get that $RC(\alpha)$ and $q_1 \circ q_1$ have the same ordering; since q_1 is a vector with positive coordinates, taking square roots preserves the ordering and the claim follows.

5.1.3 Total Communicability Centrality

Theorem: Let $G = (V, E)$ be a connected undirected graph with adjacency matrix A and e a column vector of ones. Let $TC_i(\beta) = (e^{\beta A} \cdot e)_i$ be the total communicability centrality of vertex i and TC the associated vector. Then,

- (i) $\beta \rightarrow 0^+$ implies that the ordering given by $TC(\beta)$ converges to the ordering given by $\mathbf{C}_d$; the degree centrality vector;

- (ii) $\beta \rightarrow \infty$ implies that the ordering given by $TC(\beta)$ converges to the ordering given by $\mathbf{C}_{ev}$; the eigenvector centrality vector

Proof: The Taylor expansion of function $TC_i(\beta)$ is given by:

$$TC_i(\beta) = \sum_{k=0}^{\infty} \sum_{j=1}^n \frac{\beta^k}{k!} (A^k)_{ji}$$

Note that at $k = 1$, we find the coefficient $\beta \cdot C_d(i)$, which motivates us to make a composition with:

$$f(x) = \frac{1}{\beta}(x - 1)$$

Thus, we get:

$$f(TC_i(\beta)) = C_d(i) + O(\beta)$$

With $\beta \rightarrow 0^+$, we get the claim of the first point. If we substitute the diagonal decomposition of matrix A and use the same notation as in the previous proofs, we get:

$$TC_i(\beta) = \sum_{l=1}^n \sum_{j=1}^n q_l(j) \cdot q_l(i) \cdot e^{\beta \lambda_l}$$

By scaling with $\frac{1}{e^{\beta \lambda_1}}$ and with $\beta \rightarrow \infty$, we get that the ordering given by $TC(\beta)$ is the same as the ordering given by $q_1 \cdot < q_1, e >$. Since q_1 is an eigenvector with positive components, we can get rid of the scalar product and thus obtain the eigenvector centrality C_{ev} .

5.1.4 Katz Centrality

Theorem: Let $G = (V, E)$ be a connected undirected graph with adjacency matrix A and e a column vector of ones. Let $KC_i(\alpha) = ((I - \alpha A)^{-1} \cdot e)_i$ be the Katz centrality of vertex i and KC the associated vector. Then,

- (i) $\alpha \rightarrow 0^+$ implies that the ordering given by $KC(\alpha)$ converges to the ordering given by $\mathbf{C}_d$; the degree centrality vector;
- (ii) $\alpha \rightarrow \frac{1}{\lambda_1 -}$ implies that the ordering given by $KC(\alpha)$ converges to the ordering given by $\mathbf{C}_{ev}$; the eigenvector centrality vector

Proof: The Taylor expansion of function $KC_i(\alpha)$ is given by:

$$KC_i(\alpha) = \sum_{k=0}^{\infty} \sum_{j=1}^n \alpha^k (A^k)_{ji}$$

Completely analogously to the previous proof, using the scaling function $f(x) = \frac{1}{\alpha}(x-1)$ and letting $\alpha \rightarrow 0^+$, we get the claim of the first point. Application of diagonal decomposition gives:

$$KC_i(\alpha) = \sum_{l=1}^n \sum_{j=1}^n q_l(j) \cdot q_l(i) \cdot \frac{1}{1 - \alpha \lambda_l}$$

By multiplying by $1 - \alpha \lambda_1$ and letting $\alpha \rightarrow \frac{1}{\lambda_1}^-$ (the remaining expressions cannot reach 0 in the denominator so the limit is well-defined) we get that the ordering given by $KC(\alpha)$ is the same as the ordering given by $q_1 \cdot \langle \cdot, e \rangle$. By rescaling again as in the previous proof we reduce to eigenvector centrality.

5.2 Correlation Coefficients

An important tool in comparing two results are **correlation coefficients**. The correlation of two variables gives us information about their relationship. The value of a correlation coefficient is always between -1 and 1 . A positive value indicates that an increase in one variable implies an increase in the other variable, while a negative value implies the opposite; one variable will increase and the other will decrease. Perfect correlation between variables exists when the correlation coefficient equals 1 or -1 . The closer the coefficient is to zero, the weaker the relationship between variables will be.

5.3 Kendall's Correlation Coefficient

An ideal way to compare two different variables that represent the ordering of the same set of data is to observe Kendall's correlation coefficient. It represents the difference between the probability that the same objects will be in the same order and the probability that they will be in a different order.

Specifically, let X and Y be two observations, so that pairs $(x_1, y_1), (x_2, y_2), \dots, (x_n, y_n)$ represent the ordinal numbers of the same objects in the observed orderings. Of course, the values x_i and y_i are all unique. Now, we say that pairs (x_i, y_i) and (x_j, y_j) are:

- **concordant**, if

$$\frac{x_i - x_j}{y_i - y_j} > 0.$$

This inequality is satisfied if the i -th object in both orderings X and Y is before the j -th object or is after the j -th object in both orderings.

- **discordant**, if

$$\frac{x_i - x_j}{y_i - y_j} < 0.$$

This equality is satisfied if one of the objects i, j is before the other in one ordering and after the same one in the other ordering.

- neither concordant nor discordant, if $x_i - x_j = 0$ or $y_i - y_j = 0$.

Kendall's correlation coefficient (τ) is defined as:

$$\tau = \frac{\text{No. of concordant pairs} - \text{No. of discordant pairs}}{\frac{1}{2}n(n-1)}$$

Kendall's correlation coefficient can be interpreted as follows: values of τ greater than zero indicate agreement between two different orderings, and the closer τ is to one, the greater the agreement in ordering. On the other hand, values less than zero indicate disagreement. Namely, if all pairs are concordant, then $\tau = 1$, which tells us that the data is in exactly the same ordering. If all pairs are discordant, then $\tau = -1$, which implies that the data is in completely opposite ordering (the first object in ordering X is last in ordering $Y \dots$).

6 Comparison of Measures on Data

6.1 Testing Convergence of Katz Centrality to Degree Centrality

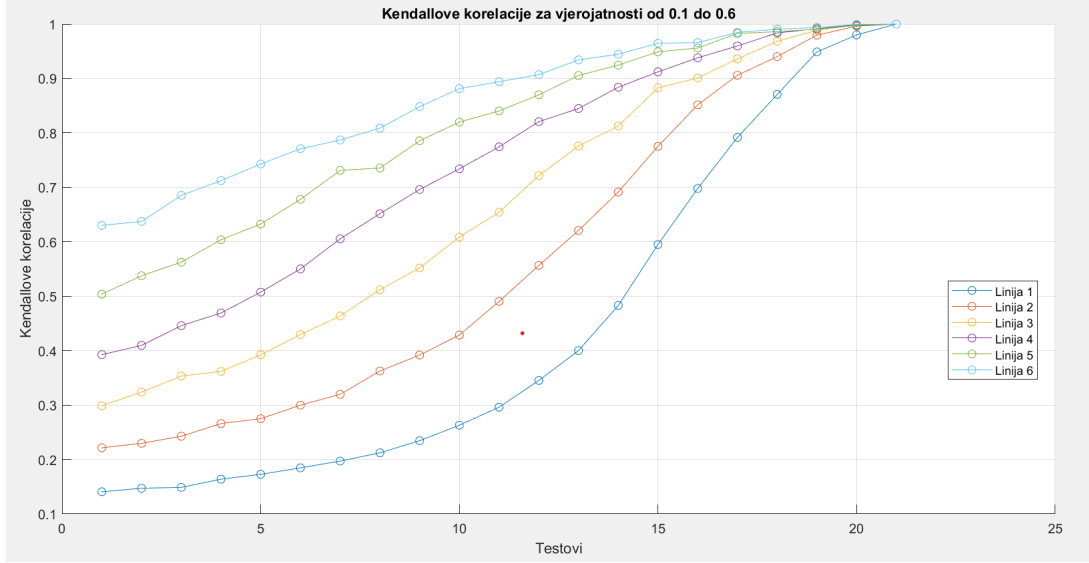


Figure 6.1: Convergence of Katz Centrality to Degree Centrality

We conduct the testing in the following way; we establish an equidistant network of 21 points placed on an interval from $\frac{\varepsilon}{2}$ to $\frac{1}{spr(A) \cdot 10^7} - \varepsilon$ where $\varepsilon = 10^{-12}$ representing coefficients α of Katz centrality. Points are arranged in descending order.

We generate one hundred Erdős–Rényi models for each of six probabilities: 0.1, 0.2, 0.3, 0.4, 0.5, 0.6. The number of vertices is fixed at 100. We compute the arithmetic mean of Kendall correlations and for each selected coefficient we record it on the graph.

Of course, as α gets closer to 0, the correlation between Katz centrality and degree centrality gets closer to 1; the same ordering of vertices is obtained.

Note that for larger probabilities in the argument of the Erdős–Rényi model, we have larger values of correlation between the two tested centralities; it is natural to conjecture that there is some proportional relationship between the number of edges in the graph (with a fixed number of vertices) and the Kendall correlation itself.

6.2 Testing Convergence of Katz Centrality to Eigenvector Centrality

We take $\varepsilon = 10^{-4}$ and coefficients α from an equidistant network of 11 points; from $\frac{\varepsilon}{2}$ to $\frac{1}{spr(A)} - \frac{\varepsilon}{2}$ in ascending order.

Table 6.1: Correlation Values for Figure 1.1

Red	$p = 0.1$	$p = 0.2$	$p = 0.3$	$p = 0.4$	$p = 0.5$	$p = 0.6$
1	0.1406	0.2215	0.2986	0.3926	0.5038	0.6301
2	0.1472	0.2296	0.3237	0.4096	0.5376	0.6373
3	0.1487	0.2427	0.3533	0.4458	0.5625	0.6852
4	0.1638	0.2661	0.3619	0.4691	0.6037	0.7123
5	0.1727	0.2749	0.3924	0.5075	0.6326	0.7429
6	0.1846	0.2998	0.4296	0.5501	0.6780	0.7710
7	0.1970	0.3197	0.4636	0.6054	0.7311	0.7869
8	0.2122	0.3625	0.5119	0.6515	0.7355	0.8088
9	0.2345	0.3920	0.5518	0.6926	0.7856	0.8484
10	0.2629	0.4286	0.6087	0.7340	0.8200	0.8814
11	0.2961	0.4905	0.6544	0.7745	0.8404	0.8941
12	0.3452	0.5565	0.7216	0.8207	0.8700	0.9071
13	0.4001	0.6208	0.7760	0.8447	0.9055	0.9342
14	0.4829	0.6915	0.8127	0.8840	0.9245	0.9446
15	0.5950	0.7753	0.8828	0.9122	0.9490	0.9648
16	0.6979	0.8516	0.9009	0.9380	0.9569	0.9659
17	0.7917	0.9060	0.9361	0.9597	0.9828	0.9848
18	0.8707	0.9403	0.9683	0.9837	0.9860	0.9904
19	0.9488	0.9797	0.9886	0.9913	0.9906	0.9938
20	0.9801	0.9961	0.9988	0.9984	0.9976	0.9991
21	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000

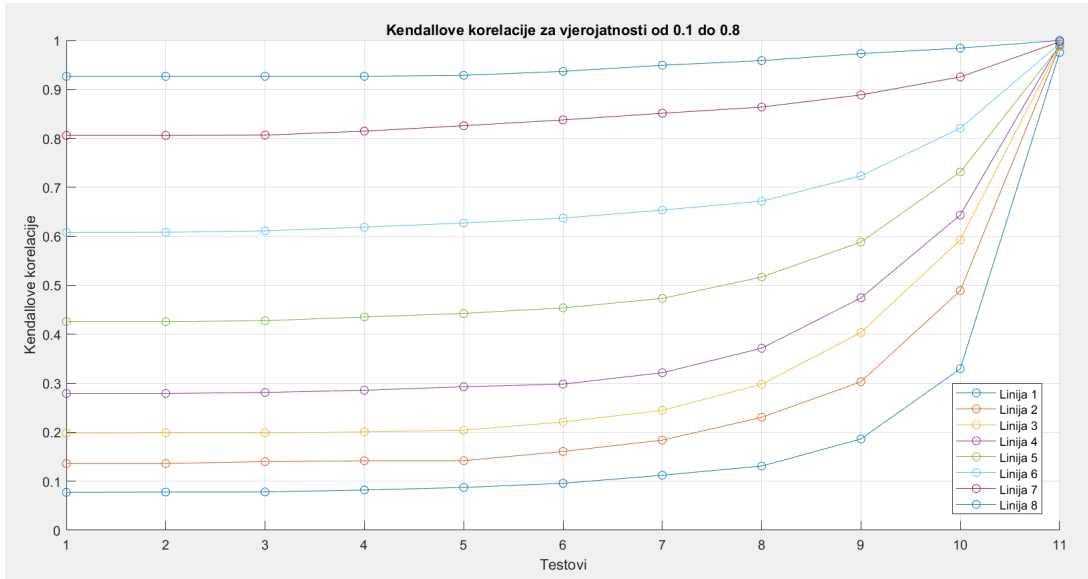


Figure 6.2: Convergence of Katz Centrality to Eigenvector Centrality

For each of the probabilities: 0.1, 0.2, 0.3, 0.4, 0.5, 0.6, 0.7, 0.8 we generate one hundred Erdős–Rényi models with fixed number of vertices of 100.

It is easy to see from the graph that the convergence of Katz centrality to eigenvector centrality has been experimentally confirmed. Furthermore, the same conjecture can be established as in the previous graph; a larger number of edges entails greater correlation.

Table 6.2: Correlation Values for Figure 1.2

Red	$p = 0.1$	$p = 0.2$	$p = 0.3$	$p = 0.4$	$p = 0.5$	$p = 0.6$	$p = 0.7$	$p = 0.8$
1	0.0773	0.1360	0.1981	0.2792	0.4261	0.6080	0.8061	0.9267
2	0.0779	0.1362	0.1991	0.2791	0.4261	0.6083	0.8060	0.9267
3	0.0781	0.1400	0.1990	0.2813	0.4277	0.6111	0.8067	0.9267
4	0.0821	0.1418	0.2008	0.2857	0.4353	0.6188	0.8148	0.9268
5	0.0873	0.1419	0.2043	0.2929	0.4426	0.6272	0.8259	0.9288
6	0.0958	0.1608	0.2210	0.2983	0.4539	0.6372	0.8377	0.9368
7	0.1122	0.1839	0.2448	0.3215	0.4733	0.6536	0.8513	0.9493
8	0.1309	0.2305	0.2981	0.3716	0.5170	0.6717	0.8639	0.9587
9	0.1863	0.3032	0.4037	0.4744	0.5883	0.7235	0.8888	0.9732
10	0.3298	0.4892	0.5930	0.6435	0.7315	0.8209	0.9256	0.9843
11	0.9754	0.9874	0.9877	0.9924	0.9882	0.9933	0.9972	0.9999

6.3 Testing Convergence of Exponential Subgraph Centrality to Eigenvector Centrality

We set the values of coefficient β in an equidistant network of 16 points on the interval from ε to $\frac{1}{10000} + \frac{\varepsilon}{2}$; in ascending order. We set $\varepsilon = 10^{-6}$.

For each of the probabilities: 0.1, 0.2, 0.3, 0.4, 0.5, 0.6 we generate one hundred models with fixed number of vertices: $n = 100$. Note that in the theorem for the given convergence we seek $\beta \rightarrow \infty$, so it is surprising that convergence was achieved so quickly (the only exception is line 1). In practice, β is limited because for sufficiently large values, computational errors arise due to finite arithmetic of computers.

Of course, as in the previous examples, in this example a proportional relationship between the number of edges and correlation is visible.

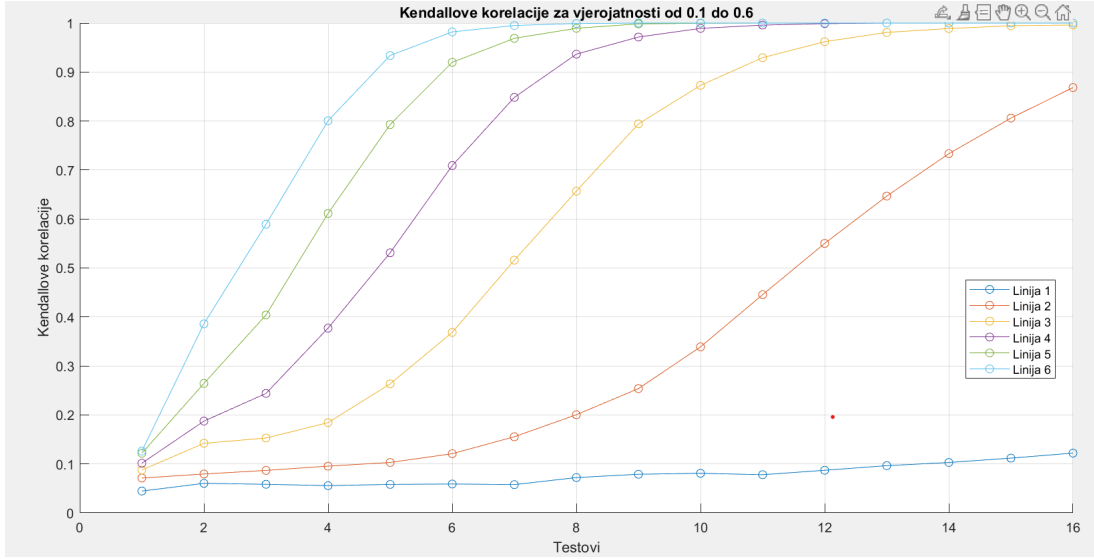


Figure 6.3: Convergence of Exponential Subgraph Centrality to Eigenvector Centrality

Table 6.3: Correlation Values for Figure 1.3

Red	$p = 0.1$	$p = 0.2$	$p = 0.3$	$p = 0.4$	$p = 0.5$	$p = 0.6$
1	0.0446	0.0711	0.0875	0.1014	0.1216	0.1260
2	0.0603	0.0794	0.1420	0.1874	0.2643	0.3860
3	0.0584	0.0867	0.1528	0.2439	0.4040	0.5891
4	0.0557	0.0930	0.1843	0.3772	0.6113	0.8005
5	0.0581	0.1030	0.2634	0.5310	0.7929	0.9300
6	0.0590	0.1209	0.3685	0.7090	0.9199	0.9820
7	0.0577	0.1554	0.5158	0.8482	0.9687	0.9954
8	0.0721	0.2005	0.6569	0.9366	0.9894	0.9991
9	0.0789	0.2538	0.7942	0.9715	0.9983	1.0000
10	0.0808	0.4455	0.8728	0.9891	0.9996	1.0000
11	0.0870	0.5501	0.9620	0.9988	1.0000	1.0000
12	0.0964	0.6467	0.9809	1.0000	1.0000	1.0000
13	0.1029	0.7334	0.9888	1.0000	1.0000	1.0000
14	0.1118	0.8059	0.9945	1.0000	1.0000	1.0000
15	0.1220	0.8683	0.9960	1.0000	1.0000	1.0000
16	0.1300	0.8971	0.9971	1.0000	1.0000	1.0000

6.4 Testing Convergence of Exponential Subgraph Centrality to Degree Centrality

In the same way as in previous examples, for each of the probabilities: 0.1, 0.2, 0.3, 0.4, 0.5, 0.6 we conduct one hundred Erdős-Rényi models with 100 vertices.

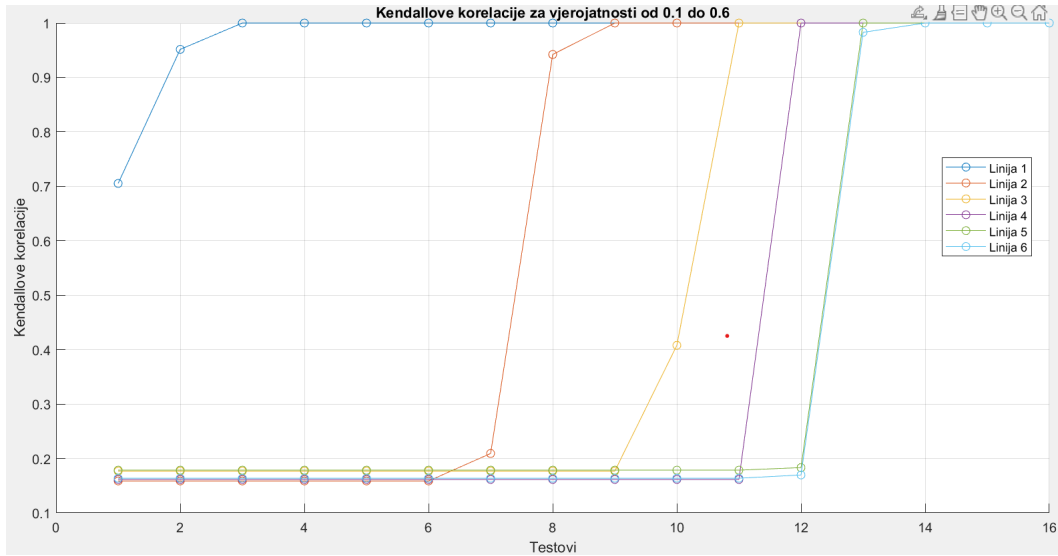


Figure 6.4: Convergence of Exponential Subgraph Centrality to Degree Centrality

We use an equidistant network to determine the coefficients of subgraph centrality; defined as in example 1.1.3., but in descending order since we want $\beta \rightarrow 0$.

We see that convergence is sharp and, unlike previous convergences, a larger number of edges implies **smaller** correlation.

Table 6.4: Correlation Values for Figure 1.4

Red	$p = 0.1$	$p = 0.2$	$p = 0.3$	$p = 0.4$	$p = 0.5$	$p = 0.6$
1	0.7053	0.1586	0.1767	0.1615	0.1785	0.1637
2	0.9516	0.1586	0.1767	0.1615	0.1785	0.1637
3	1.0000	0.1586	0.1767	0.1615	0.1785	0.1637
4	1.0000	0.1586	0.1767	0.1615	0.1785	0.1637
5	1.0000	0.1586	0.1767	0.1615	0.1785	0.1637
6	1.0000	0.1586	0.1767	0.1615	0.1785	0.1637
7	1.0000	0.2092	0.1767	0.1615	0.1785	0.1637
8	1.0000	0.9423	0.1767	0.1615	0.1785	0.1637
9	1.0000	1.0000	0.1767	0.1615	0.1785	0.1637
10	1.0000	1.0000	0.4078	1.0000	1.0000	0.1832
11	1.0000	1.0000	1.0000	1.0000	1.0000	0.9829
12	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
13	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
14	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
15	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
16	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000

6.5 Comparison of Betweenness Centrality and Closeness Centrality

So that some measures do not escape the analysis, we will compare the distance centrality of a vertex and the betweenness centrality on generation models. The results are given in the following table:

p	0.50	0.55	0.60	0.65	0.70	0.75	0.80	0.85	0.90	0.95
ErdosRenyi	0.076	0.076	0.090	0.004	0.079	0.062	0.026	0.077	0.030	0.060
Watts-Strogatz	0.016	-0.001	-0.008	0.027	0.019	0.049	-0.020	0.050	0.031	0.024
Barabási - Albert	0.526	0.508	0.576	0.592	0.613	0.594	0.577	0.552	0.581	0.536

Data were obtained with 100 vertices; for each probability p the graph was generated 10 times, and the final correlation value is the arithmetic mean of all 10 correlations. The first two models practically show no signs of affecting the correlation between the two centralities, but the last model takes much larger values.

6.6 Application of Centrality Measures

Now it is time to test the theoretically discussed methods on real data. As already mentioned, finding central vertices is necessary in all sorts of situations. The most influential people on social networks, individuals responsible for disease transmission among populations are just some examples where it is useful to learn about the most important individuals.

As the first example, let us look at the large network of people at one large *European research institute*. The graph consists of 1005 vertices representing employees at that institute and we observe communication between them. An edge (u, v) is found in the graph if person u sent an email to person v . The goal is to examine which person is at the top of the ordering obtained by some centrality measures and to see if different measures give similar results.

In Table 6.5 we can see the top 20 vertices with respect to degree centrality (DC), closeness centrality (CC) and eigenvector centrality (EC). It is immediately clear that first place in the ordering of all tested measures is occupied by vertex 161, which tells us that this person received the most emails, has the most influential neighbors and has the smallest distance from all other employees. Besides that, we see that vertices 108, 63, 184 and 122 are all also at the top regardless of the measure, which also tells us much about the influence of those employees.

For the director of the institute, such information would be useful if he wanted to spread important news throughout the institute in the shortest possible time or if he wanted to learn the experiences of employees with the most contact with people.

We can see in Table 6.6 Kendall coefficients of the observed measures. Eigenvector centrality and degree centrality are the most closely related, and eigenvector centrality and closeness centrality are the least related. However, we do not observe great correlation between any pair of measures, which is intuitively clear since different measures give different information.

CC	DC	EC
161	161	161
63	803	108
108	83	63
122	122	435
87	108	122
435	87	184
184	63	129
130	980	257
65	250	250
129	435	130
107	993	83
167	756	284
6	184	115
83	6	88
284	14	107
212	167	167
257	130	212
302	212	106
106	88	213
250	106	170

Table 6.5: Top 20 vertices

Table 6.6: Kendall Coefficient of Measures

Comparison of Measures	Kendall Coefficient
$\tau(\text{DC}, \text{CC})$	0.3021
$\tau(\text{EC}, \text{CC})$	0.2718
$\tau(\text{DC}, \text{EC})$	0.3226

7 Conclusion

In this paper, centrality measures such as vertex degree, closeness centrality, eigenvector centrality, Katz centrality, and betweenness centrality were investigated.

We theoretically investigated the relationship between Katz centrality and degree centrality as well as eigenvector centrality on the Erdős–Rényi graph. It was proven that the ordering obtained by Katz centrality tends to degree centrality when the coefficient α tends to 0. When α tends to $\frac{1}{\lambda}$, Katz centrality behaves like eigenvector centrality.

We also experimentally showed that we can conjecture all the claims of theoretical conclu-

sions about Katz centrality.

We came to similar conclusions theoretically and experimentally when observing exponential subgraph centralities, but here the parameter p with which the graph is generated is also crucial. Additionally, we experimentally showed a relationship between the probabilistic variable in the Erdős–Rényi model and Kendall correlation for cases of Katz centrality with eigenvector centrality and degree centrality; analogously for exponential subgraph.

Similarly, we theoretically showed that exponential subgraph centrality converges to degree centrality when the coefficient β tends to 0 and when β tends to infinity, the observed centrality tends to eigenvector centrality. We came to the same conclusions when examining the coefficients in total communicability centrality.

We investigated the effect of graph model on the relationship between betweenness centrality and closeness centrality. It was noted that different models give different results, precisely the Barabási–Albert model generates graphs on which the average Kendall coefficient is between 0.5 and 0.6, which indicates a large overlap in vertex ordering. Other models generate graphs whose vertices differ significantly in orderings obtained by different centralities. We conclude that when a graph has vertices that stand out significantly by the number of neighbors, those vertices will be of great importance in many other measures, such as the distance measure from all vertices or participation in most paths.

On real data from an institute, we tested centrality measures and found relatively weak, but existing relationship of various centralities. Measuring centrality can be useful in large networks if we want to spread news in the shortest time, so we would find the most influential vertices using the closeness centrality measure.

In conclusion, centrality measures are useful in studying large networks if we want to learn about the most influential and important vertices. It is good to understand the application of various centrality measures because, depending on the parameters, one can find the perfect recipe for any requirement, thus enabling optimized decision-making and better understanding of network structure.

8 References

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